



One line stops. **Three verified routes.** One human decides.

Day 2 adds one local learned suggestion - never a generated route.

01 · SIMULATED TWIN

SYNTHETIC SCENARIO

Simulated twin

LINE 2 STOP · 10:00-14:00

L1

L2

L3

SIMULATED TELEMETRY

L2 speed **0.0 m/min**

Temperature **cooling**

Seeded · deterministic · clearly synthetic

3 lines · 10 orders

Twin view mirrors the same executable state.

02 · NON-LEARNING PLANNER

DETERMINISTIC · NON-LEARNING

Constraint planner

VALIDATE
Constraints

BUILD
3 routes

SCORE
KPIs

HARD CONSTRAINTS

Compatibility

No overlap

Stop window

Horizon

Builds routes · no training · no LLM

SEPARATE EXACT BOUNDED VERIFIER

shared validation + KPI functions

17,856

→

10,440

→

3

assignment / sequence candidates

feasible schedules

unique policy optima

Does not reuse planner construction logic; exhaustively cross-checks the 3 policy optima.

03 · ROUTES + LOCAL ADVISOR

THREE PLANNER-BUILT ROUTES

SERVICE

140 min delay

COST

1,931.45 DT*

SUGGESTED

STABILITY

0 moves

SEPARATE LOCAL LEARNED RECOMMENDER

Suggests **COST**

Trained on **687** incidents simulated by the twin **93.6%** held-out synthetic test accuracy

Selects among existing routes · never creates or edits a schedule.

*illustrative synthetic comparison output

04 · HUMAN AUTHORITY

HUMAN IN COMMAND



Production manager

- 1 Preview twin + Gantt
- 2 Review routes + suggestion
- 3 Approve or reject

NO MACHINE COMMAND

In-memory approval + timestamped audit event

Suggestion is advisory; manager may choose any route.

LOCAL DEPLOYMENT + EVIDENCE BOUNDARY

SYNTHETIC DATA

SIMULATED TELEMETRY

LOCAL + OFFLINE

READ-ONLY

NO MACHINE CONTROL

9/9 CORE

5/5 ML

twin-engine.js → 3 routes
recommender.js → suggestion only

93.6% is held-out synthetic validation, not plant performance.
Pilot rule: retrain locally on the site's real incident history.